CloudNine Systems

Group 4 | Milestone 1 | ERD

23 core tables. Four work packages. One connected logical design.

Core workflow

Client requests -> reservation and items -> deployment and allocation

Provisioned service -> usage -> charge -> payer share -> invoice

A

Member A: Clients and reservations

6 tables | Detailed view: page 3

B

Member B: Infrastructure and availability

6 tables | Detailed view: page 4

C

Member C: Deployments and services

5 tables | Detailed view: page 5

D

Member D: Metering and billing

6 tables | Detailed view: page 6

Page 2: all 23 entities, primary keys and principal domain relationships.

Pages 3-6: every owned attribute and every FK, including actors and self-links.

Page 7: composite uniqueness and important constraints.

1 = exactly one; 0..1 = optional one; 0..* = zero or more.

PK = primary key; FK = foreign key; UK = single-column unique key.

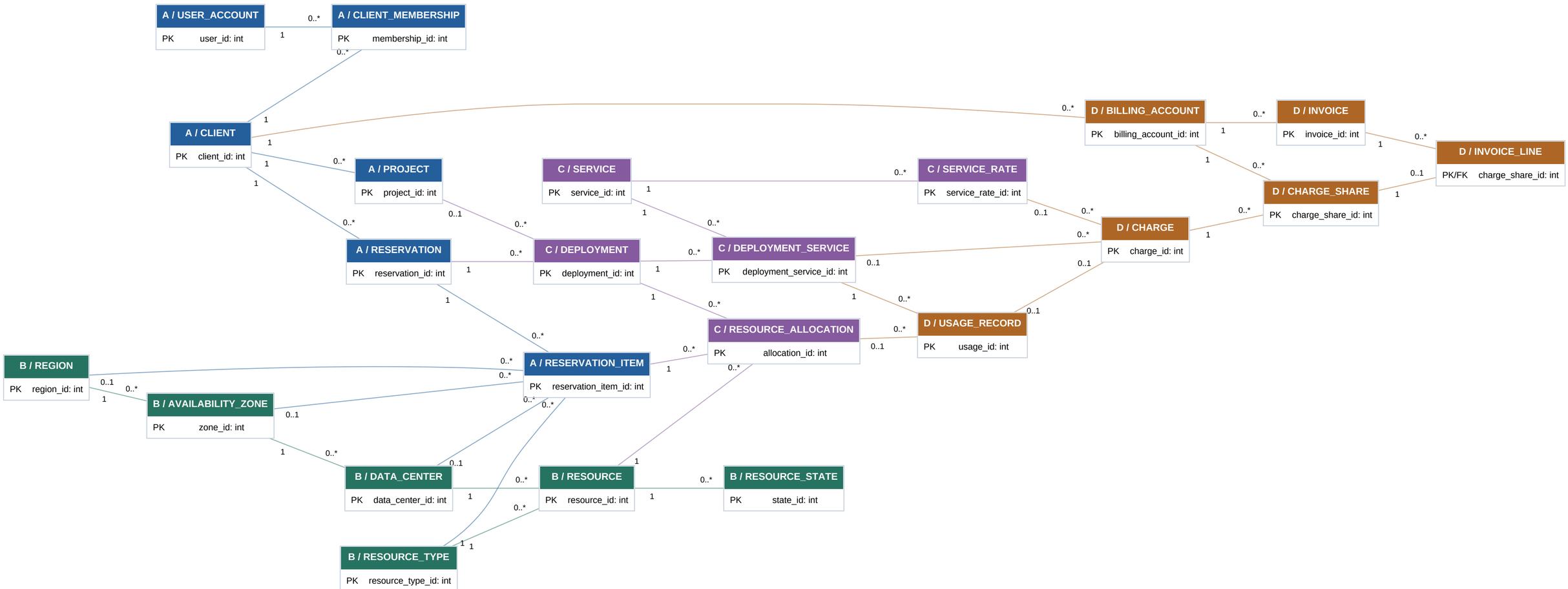
? = nullable; gray boxes = external references, not duplicate tables.

All money is USD; all timestamps are UTC.

3NF reasoning and scope assumptions are in the draft analysis report.

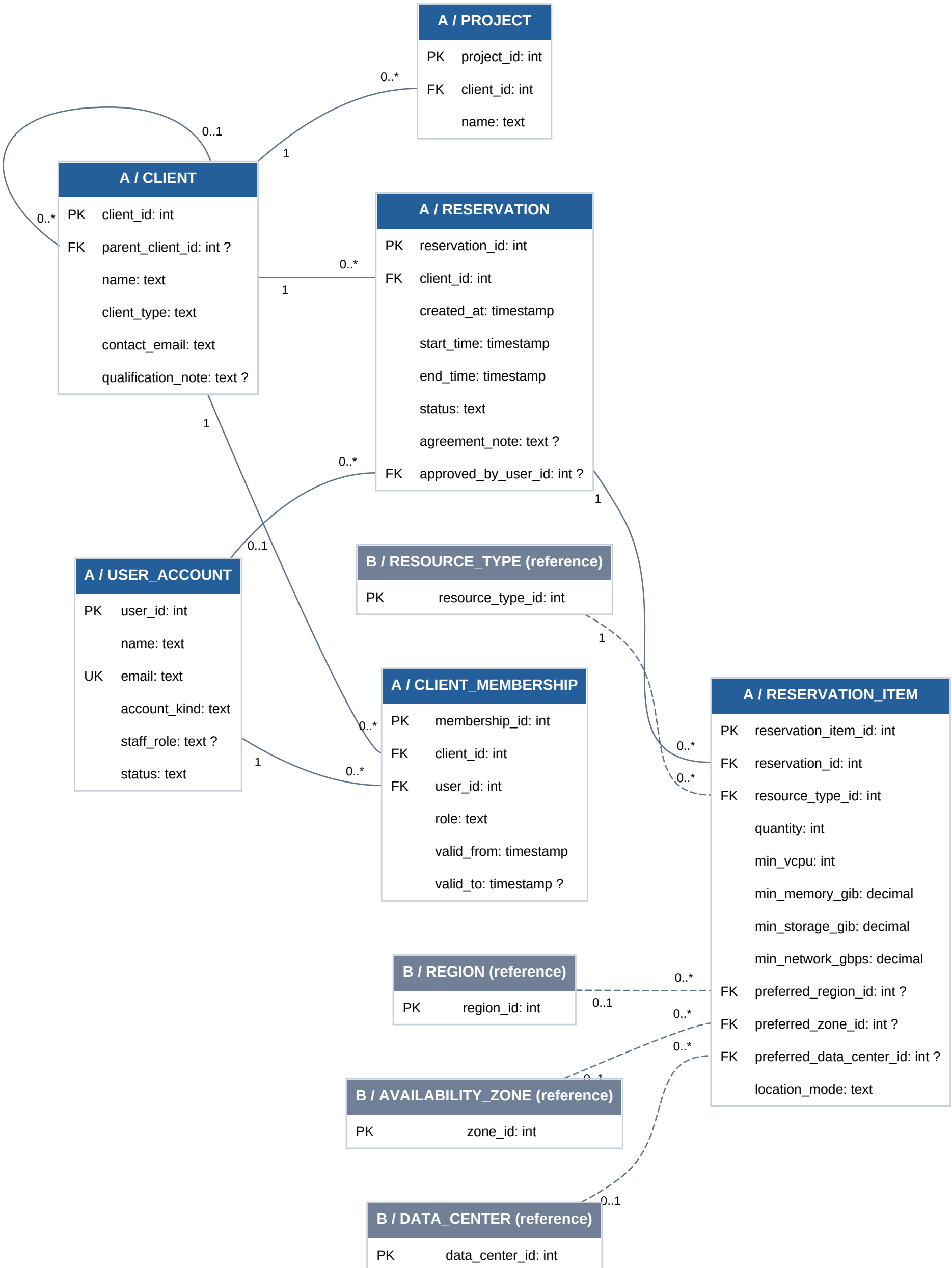
All-table overview | 23 entities

A = clients / requests; B = resources; C = deployments / services; D = usage / billing. Actor and recursive FKs appear in detail views.



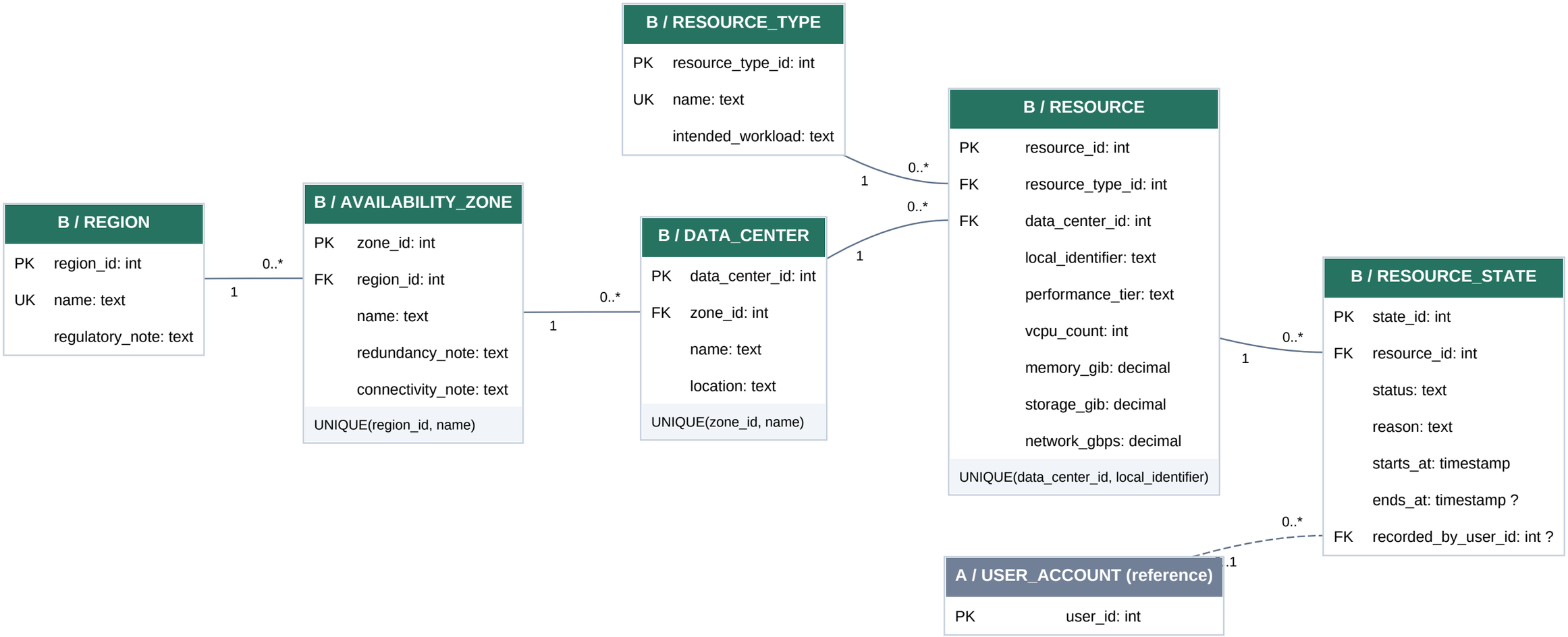
Member A | Clients and reservations

6 owned tables. Gray boxes reference other owners. Lines connect parent PKs to child FKs.



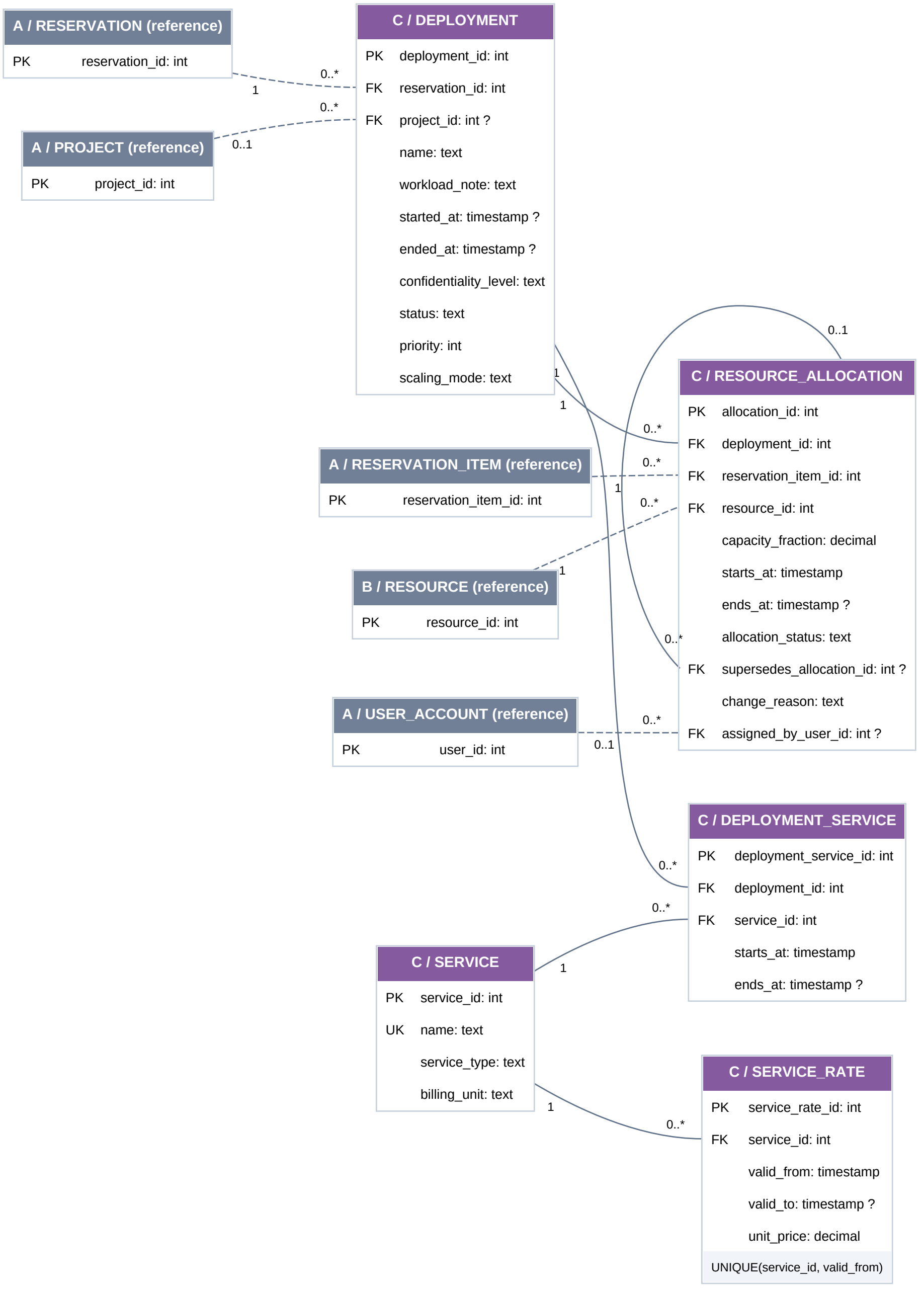
Member B | Infrastructure and availability

6 owned tables. Gray boxes reference other owners. Lines connect parent PKs to child FKs.



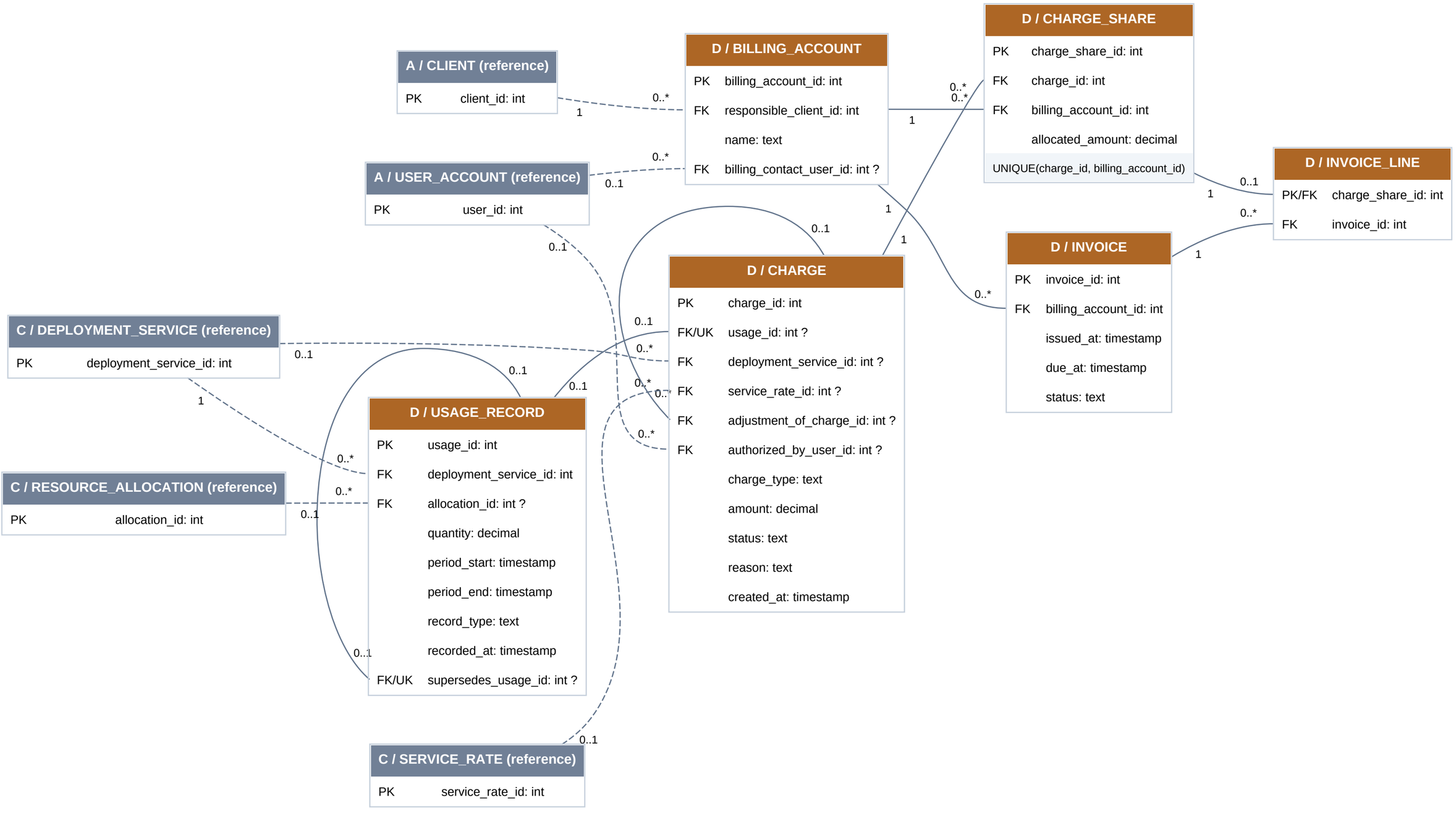
Member C | Deployments and services

5 owned tables. Gray boxes reference other owners. Lines connect parent PKs to child FKs.



Member D | Metering and billing

6 owned tables. Gray boxes reference other owners. Lines connect parent PKs to child FKs.



Keys and integrity rules

Composite UNIQUE constraints

AVAILABILITY_ZONE: UNIQUE(region_id, name)

DATA_CENTER: UNIQUE(zone_id, name)

RESOURCE: UNIQUE(data_center_id, local_identifier)

SERVICE_RATE: UNIQUE(service_id, valid_from)

CHARGE_SHARE: UNIQUE(charge_id, billing_account_id)

Exclusive choices (XOR)

RESERVATION_ITEM: at most one preferred location level.

CHARGE: exactly one of usage_id, deployment_service_id,

or adjustment_of_charge_id. Metered charges require service_rate_id.

Lifecycle and financial rules

Approved reservations contain items; issued invoices contain lines.

Resource type, minimum capacity, dates and mandatory locations must match.

Concurrent assigned fractions cannot exceed one per resource.

Concurrent item assignments cannot exceed the requested instance count.

State intervals for the same resource cannot overlap.

Final shares total the charge; posted charges and shares retain history.

An invoice line must reference a final share for the same payer account.

A share appears on at most one invoice (INVOICE_LINE PK/FK).

History and normalization

Allocation, availability, rate and usage-correction history are retained.

Configuration lineage, reservation revisions and full event logs are deferred.

Do not repeat region or zone in RESOURCE, or deployment in USAGE_RECORD.

Do not repeat invoice membership or a billing rule in CHARGE_SHARE.

Temporal, cross-table and aggregate rules need transactional checks.

See analysis sections 3-5 for dependencies, assumptions and limitations.